

Problem Set 4

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Question 1

Instructions

We're interested in modeling the historical causes of child mortality. We have data from 26855 children born in Skellefteå, Sweden from 1850 to 1884. Using the "child" dataset in the `eha` library, fit a Cox Proportional Hazard model using mother's age and infant's gender as covariates. Present and interpret the output.

Answers

To answer this question, a Cox proportional hazards model was fitted to examine the risk of death before age 15. The included variables are maternal age at birth and the child's sex. The results are shown in Table 1 below.

Table 1 reports the estimates from a Cox proportional hazards model of mortality before age 15, based on 26,574 children, of whom 5,616 experienced the event of interest (death before age 15). The coefficient on mother's age is 0.008 with a standard error of 0.002, and it is statistically significant at the 0.1% level ($p < 0.001$). Because Cox coefficients are reported in log-hazard terms, this implies that a one-year increase in maternal age increases the hazard of death by approximately $e^{0.008} - 1 \approx 0.8\%$, holding the child's sex constant. This is a small effect in magnitude, but interpreted over a larger age gap, the effect becomes more noticeable: a 5-year increase in maternal age at birth corresponds to an estimated hazard ratio of about 1.039, or a 3.9% higher mortality hazard, while a 10-year increase corresponds to a hazard ratio of about 1.079, or a 7.9% higher mortality hazard.

The coefficient on *Female* is -0.082 with a standard error of 0.027, and this effect is statistically significant at the 1% level ($p < 0.01$). Since males are the omitted reference category, this coefficient compares girls to boys. Converting the coefficient

Table 1: Cox Proportional Hazards Model for Child Mortality

	<i>Dependent variable:</i>
	Hazard of death before age 15
Mother's age	0.008*** (0.002)
Female	-0.082** (0.027)
Observations	26,574
R ²	0.001
Max. Possible R ²	0.986
Log Likelihood	-56,503.480
Wald Test	22.520*** (df = 2)
LR Test	22.518*** (df = 2)
Score (Logrank) Test	22.530*** (df = 2)
<i>Note:</i>	*p<0.05; **p<0.01; ***p<0.001

into a hazard ratio gives $e^{-0.082} \approx 0.921$. Thus, female children have an estimated mortality hazard that is about 7.9% lower than that of male children, holding maternal age constant. Equivalently, male children face a hazard about $1/0.921 \approx 1.086$, or 8.6%, higher than female children. This indicates a statistically meaningful gender difference in survival, with girls exhibiting a lower risk of death before age 15.

The model as a whole is statistically significant. The Wald test statistic is 22.520 on 2 degrees of freedom, the likelihood ratio test statistic is 22.518 on 2 degrees of freedom, and the score (logrank) test statistic is 22.530 on 2 degrees of freedom; all three are significant at the 0.1% level. This means that the null hypothesis that both coefficients are jointly equal to zero can be rejected very strongly. In other words, taken together, maternal age and child sex are significantly related to child mortality risk in this sample.

At the same time, the substantive explanatory power of the model is limited. The reported pseudo- R^2 is 0.001, indicating very limited explanatory power. This is consistent with the concordance statistic (approximately 0.519), which is only slightly above 0.5, the value expected under random prediction. Substantively, this suggests that although maternal age and child sex have precisely estimated and statistically significant effects, they do not by themselves account for much of the overall het-

erogeneity in mortality risk. Other factors, such as household conditions, legitimacy status, socioeconomic background, and environmental circumstances, are likely to play a substantial role.

All of this was done using the code below:

```
1 #### Question 1
2
3 library(eha)
4 library(survival)
5 library(stargazer)
6
7 # Load dataset
8 data(child)
9
10 # Inspect data
11 head(child)
12 str(child)
13 summary(child)
14
15 # Fit Cox proportional hazards model
16 cox_model <- coxph(Surv(enter, exit, event) ~ m.age + sex, data = child
17 )
18 summary(cox_model)
19
20 stargazer(
21   cox_model,
22   type = "latex",
23   title = "Cox Proportional Hazards Model for Child Mortality",
24   dep.var.labels = "Hazard of death before age 15",
25   covariate.labels = c("Mother's age", "Female"),
26   digits = 3,
27   no.space = TRUE,
28   star.cutoffs = c(0.05, 0.01, 0.001),
29   out = "cox_model_table.tex")
```

Question 2

Instructions

We want to estimate how the amount of damage caused by a disaster relates to (1) whether disaster relief is provided and (2) the amount of relief that is provided conditional on a donation occurring when a disaster hits a given country. Estimate a Heckman selection model using the `disasters_response` data on GitHub in which `binContribution` is the outcome for the selection equation, and `originalContri-`

butionMillionUSDLogged is the outcome for the second equation. Use occurrences + deathsEM + normalizedDamageEMLogged as your input variables for both equations. Present and interpret the output.

Answers

To answer this question, a Heckman selection model was fitted to examine both the likelihood of disaster relief being provided and the amount of relief given, conditional on a contribution occurring. This approach is appropriate because the outcome variable aid amount is only observed for cases where a donation is made, leading to a non-random sample and potential selection bias. The Heckman model corrects for this by jointly estimating a selection equation for whether aid is given and an outcome equation for the amount of aid. The results are shown in Table 2 below.

Table 2: Heckman Selection Model for Disaster Aid

	<i>Dependent variable:</i>
	originalContributionMillionUSDLogged
occurrences	−0.002 (0.006)
deathsEM	0.006*** (0.002)
normalizedDamageEMLogged	−0.018*** (0.005)
Constant	0.413 (0.407)
Observations	49,096
Log Likelihood	−17,557.780
ρ	−0.527*** (0.091)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

The coefficient on occurrences is negative (−0.002) and statistically insignificant, indicating that the number of disasters has no meaningful effect on the amount of aid provided once a donation is made. This suggests that while more frequent

disasters may influence the likelihood of receiving aid, they do not affect the size of contributions conditional on aid being given.

The coefficient on `deathsEM` is positive (0.006) and statistically significant at the 1% level. This implies that more deadly disasters are associated with higher levels of aid. Specifically, a one-unit increase in `deathsEM` is associated with an approximate 0.6% increase in the amount of aid provided, holding other factors constant. This shows that donors respond more strongly to the human impact of disasters when determining aid amounts.

In contrast, `normalizedDamageEMLogged` has a negative and statistically significant coefficient (−0.018). This implies that greater economic damage is associated with lower aid amounts, conditional on aid being given. Interpreting the coefficient in percentage terms, a one-unit increase in logged economic damage is associated with approximately a 1.8% decrease in aid. This counterintuitive result may reflect that economic damage does not drive aid allocation to the same extent as human losses, or that aid is spread more thinly across severely affected contexts.

The estimated correlation coefficient between the error terms, $\rho = -0.527$, is negative and statistically significant at the 1% level. This provides strong evidence of selection bias and justifies the use of the Heckman model. The negative sign indicates that unobserved factors that increase the likelihood of providing aid are associated with lower aid amounts. In other words, countries that are more likely to give aid for unobserved reasons tend to provide smaller contributions, while less frequent donors tend to give larger amounts when they do contribute.

Overall, the results suggest that while disaster severity increases the likelihood of receiving aid, the determinants of aid amounts differ, with human impact (deaths) playing a more important role than economic damage, and with clear evidence of non-random selection into aid provision.

All of this was done using the code below:

```
1 ### Question 2
2 setwd("C:/Users/onnos/Documents/GitHub/StatsII_2026/problemSets/PS04/my
   _answers")
3
4 # load dataset
5 disaster_response <- read.csv("disaster_response.csv")
6
7 # Inspect
8 head(disaster_response)
9 str(disaster_response)
10 summary(disaster_response)
```

```

11
12 library(sampleSelection)
13
14 # Heckman selection model
15 heckman_model <- selection(
16   selection = binContribution ~ occurrences + deathsEM +
17     normalizedDamageEMLogged,
18   outcome   = originalContributionMillionUSDLogged ~ occurrences +
19     deathsEM + normalizedDamageEMLogged,
20   data = disaster_response,
21   method = "ml")
22
23 # View results
24 summary(heckman_model)
25
26 stargazer(
27   heckman_model,
28   type = "latex",
29   title = "Heckman Selection Model for Disaster Aid",
30   out = "heckman_table.tex")

```